

Particle Methods

Many success stories in high-dimensional inference!

Sampling & MCMC:

- Langevin algorithm (unadjusted, metropolis adjusted, underdamped, ...)

- Forward-Backward diffusion, DDPM ([Song, Ermon 2025](#), [Ho et al. 2020](#))

- SVGD ([Liu, Wang 2016](#), [Liu 2017](#))

Mixture Modeling:

- Variational Inference ([Lambert et al. 2022](#))

- Approximate NPMLE ([Yan et al. 2023](#))

Filtering, Numerical PDE:

- Ensemble/Particle Kalman Filtering, Sequential MCMC ([Doucet et al. 2001](#))

- Smoothed Particle Hydrodynamics ([J. Monaghan, 2005](#))

- Measure Quantization ([Xu et al, 2022](#), [Belhadji 2025](#))

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Computer Graphics...?